

RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering

Academic Year: 2019 - 2020 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE-‘A’

Course Code & Title: EC8393 & Fundamentals of Data Structures in C

Practice's Followed: Simulation of sorting using Virtual lab

Type of Activity: Virtual lab

Topic: Insertion Sort, Merge Sort, Quick Sort

Date: 11-10-19

Objectives:

O1: To simulate various sorting algorithms with various inputs, including best case, worst case, and average inputs

O2: To analyze the performance of various sorting such as Insertion Sort, Merge Sort, Quick Sort.

Justification for choosing the particular topic for simulation:

Sorting is the main topic in V unit. To make the students familiar with various sorting algorithm I planned to give hands on simulation practices. One of the primary advantages of choosing simulation is that to provide students with practical experiences when dealing with various time complexities.

Virtual Laboratory:

The Virtual Laboratory is an interactive environment for creating and conducting simulated experiments: a playground for experimentation. A virtual laboratory is an on-screen simulator or calculator that learners use to test ideas and observe results. They allow the learner to “tinker” with laboratory equipment that behaves in almost the same way as it would in a real environment. Learners perform a series of experiments that yield authentic results.

Instructions to perform a simulation experiment:

Prerequisites to run the simulation:

For Windows

1) Go to <http://get.adobe.com/flashplayer/> and select the version of flash player suitable for your system. The Flash Player installer should prompt to begin installation after the download is complete. If it does not, start the installer directly from Downloads folder.

2) After downloading Flash Player, close all open browsers.

3) Locate the install flash player file. Unless you have specified otherwise, the file is saved in your Downloads folder.

4) Double-click the installer file to begin the Flash Player installation.

Virtual lab Link:

The following link is used to access the virtual lab.

1. Go to <http://vlab.co.in/> //URL for Virtual lab

2. Choose Computer Science and Engineering

3. This page contains broad area of virtual labs

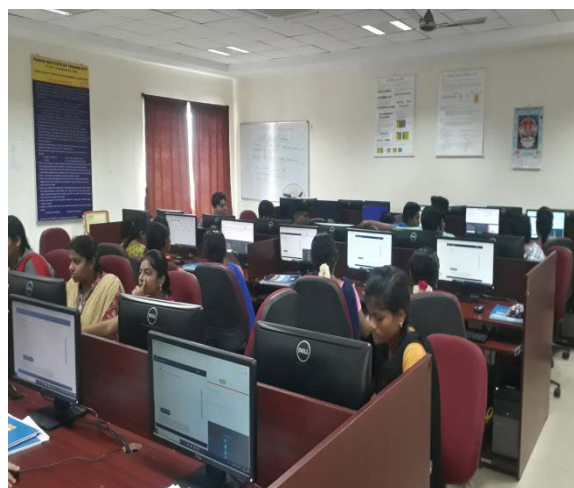
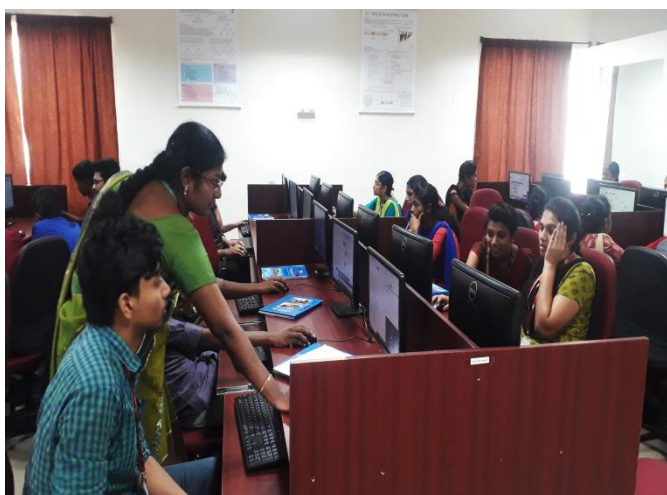
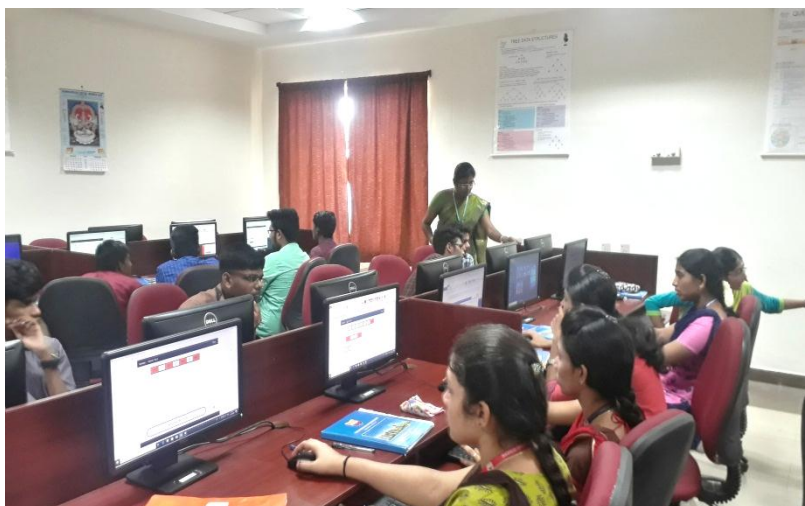
4. Select Data Structures lab from the listed.

5. Choose List of Experiments from the drop down list.

6. Select sorting using arrays and then choose simulation button.

Activity Description:

Sorting algorithms were discussed in class room. For better understanding of various sorting algorithms, students were given hands on experience on sorting using virtual lab. The activity is conducted on CC-III lab. Each student practiced simulation individually for sorting algorithm such as merge sort, insertion sort and quick sort. Finally their outputs for different cases were discussed.

**Reflection critique****Post Implementation:**

- Based on the observation, students worked joyfully because of visualization blow in their mind.
- Students have got clear fundamental knowledge on Insertion Sort, Merge Sort, Quick Sort.
- Some of the students asked doubts on simulation set up and time complexity of different sorting.

Relevance of Program Outcomes

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
O1	3	3	2	2	2					1		1
O2	3	3	2	2	2					1		1

References

Visit to the following URL to perform a simulation for sorting

<http://cse01-iiith.vlabs.ac.in/exp3/quick.html>

<http://cse01-iiith.vlabs.ac.in/exp3/merge.html>

<http://cse01-iiith.vlabs.ac.in/exp3/insertion.html>

Faculty-Incharge

S.Manjula, AP/CSE

HOD

Dr.K.Vijayalakshmi